

- 1. Which component makes up the largest percentage of total blood volume?**
 - a. Erythrocytes
 - b. Plasma
 - c. Leukocytes
 - d. Thrombocytes
- 2. What is the normal pH range of human blood?**
 - a. 6.55 – 6.65
 - b. 7.15 – 7.25
 - c. 7.35 – 7.4
 - d. 7.55 – 7.65
- 3. Which organ synthesizes the vast majority of plasma proteins, excluding immunoglobulins?**
 - a. Spleen
 - b. Red bone marrow
 - c. Kidneys
 - d. Liver
- 4. Which plasma protein is primarily responsible for maintaining the colloid osmotic (oncotic) pressure of blood?**
 - a. Albumin
 - b. Fibrinogen
 - c. Gamma globulin
 - d. Prothrombin
- 5. What is the normal average concentration of total plasma proteins in a healthy adult?**
 - a. 3 – 5 g/dL
 - b. 6 – 8 g/dL
 - c. 9 – 11 g/dL
 - d. 12 – 14 g/dL
- 6. Blood serum differs from blood plasma because serum lacks which of the following?**
 - a. Albumin
 - b. Immunoglobulins
 - c. Clotting factors
 - d. Electrolytes

- 7. Which globulin fraction contains the antibodies responsible for the humoral immune defense system?**
- a. Alpha-1 globulin
 - b. Alpha-2 globulin
 - c. Beta globulin
 - d. Gamma globulin
- 8. Which plasma protein has a fibrous structure and plays a critical role in the final stage of blood coagulation?**
- a. Transferrin
 - b. Fibrinogen
 - c. Regulatory hormone
 - d. Prealbumin
- 9. Plasma is composed of approximately what percentage of water?**
- a. 45%
 - b. 70%
 - c. 90%
 - d. 99%
- 10. Which plasma protein is responsible for transporting hormones, iron and CO₂?**
- a. Albumin
 - b. Prothrombin
 - c. Fibrinogen
 - d. Alpha-1-antitrypsin
- 11. Edema of liver failure starts at which of the following sites?**
- a. Preorbital around the eyes.
 - b. Abdominal in the peritoneal cavity (ascites).
 - c. In the lower limbs.
 - d. In the upper limbs.